

Computer Vision:

Week-1 (12-16 Jan) Theory Lectures

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Camera calibration

- It is the **process** of estimating a camera's **geometric parameters** so you can accurately map 3D points in the real world to 2D pixels in an image (and often back again). It's what lets you make **measurements from images**, do 3D reconstruction, AR overlays, accurate pose estimation, etc.

What calibration estimates

Intrinsic parameters (inside the camera)

- These describe how the camera forms an image
- Focal lengths f_x, f_y
- Principal point: C_x, C_y

-Lens distortion coefficients:

Radial distortion (barrel/pincushion): $k_1, k_2, k_3, \dots$

Tangential distortion (lens not perfectly aligned): $p_1, p_2, \dots$

camera intrinsic matrix, typically denoted as K .

$$K = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix}$$

Extrinsic parameters (camera pose)

These describe where the camera is relative to the world:

- Rotation R
- Translation t

How calibration is usually done :

- Take multiple images of a known pattern (commonly a checkerboard).
- Detect feature points (checkerboard corners).
- Solve an optimization problem (often minimizing reprojection error):.

Outputs you get:

- Intrinsic K .
- Distortion coefficients
- Pose (R, t) for each calibration image (if you calibrate with multiple views)
- Sometimes a way to undistort images (remove lens distortion).

Why it matters:

Without calibration, many CV tasks become inaccurate:

- 3D reconstruction
- Depth estimation from stereo
- Measuring distances/angles from images
- Robot navigation and grasping

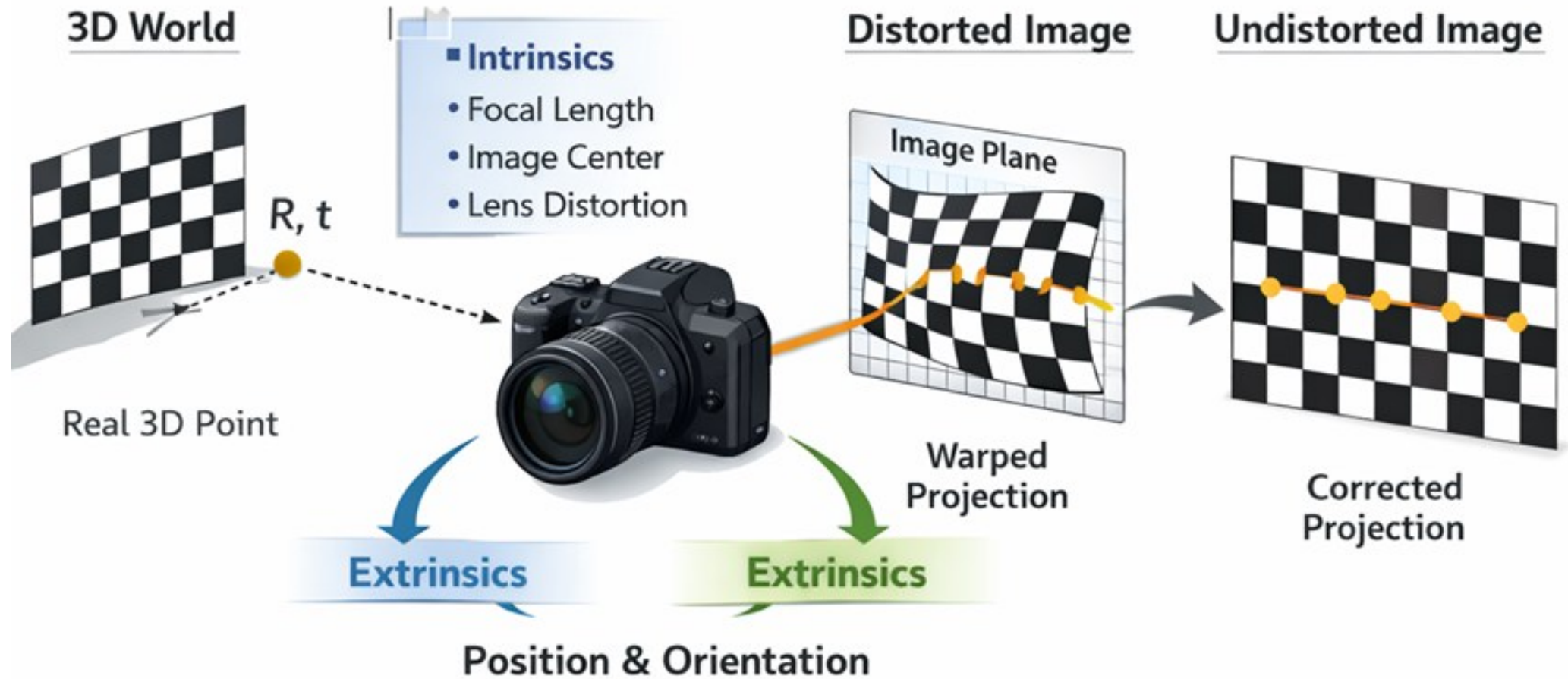
Camera calibration is basically teaching a computer what your camera “looks like” mathematically:

- It finds the camera’s **internal settings** (like how strong the zoom is and where the image center is) and how much the lens warps the image (distortion).
- It can also find the **camera’s position** and orientation (where it is and which way it’s pointing) relative to the scene.

Camera calibration is basically teaching a computer what your camera “looks like” mathematically:

- We usually do it by taking photos of a known **pattern** (like a checkerboard) and adjusting the camera parameters until the projected points match the real points in the images.

Camera Calibration



Homographies (The "Plane Warper")

- A **Homography** is a mathematical transformation that maps points from one flat surface (plane) to another.
- Think of it as the "perspective correction" tool. If you look at a poster on a wall from a sharp angle, it looks distorted.
- A homography calculates how to **warp** that image so it looks like you are standing directly in front of it.

Real-World Applications::

- **Document Scanners:** Apps like CamScanner use homography to flatten a picture of a receipt taken from an angle.
- **Panoramas:** Stitching multiple images together. Since the camera usually just rotates, we can use homographies to align the overlapping sections perfectly.
- **Virtual Advertising:** Projecting a logo onto a sports field TV broadcast so it looks like it's painted on the grass.

Multiple View Geometry:

- When we have **two or more** images of the same scene taken from different positions (like our two eyes), we can calculate 3D depth.
- This is the geometry of structure from motion.
- While Homography deals with planes, Multiple View Geometry deals with 3D space.

Key Mechanics:

- **Epipolar Geometry:** If we see a point in the Left Image, we don't have to search the entire Right Image to find its match.
- Geometry tells us the match must lie on a specific line called the "Epipolar Line." This
- drastically speeds up computer processing.

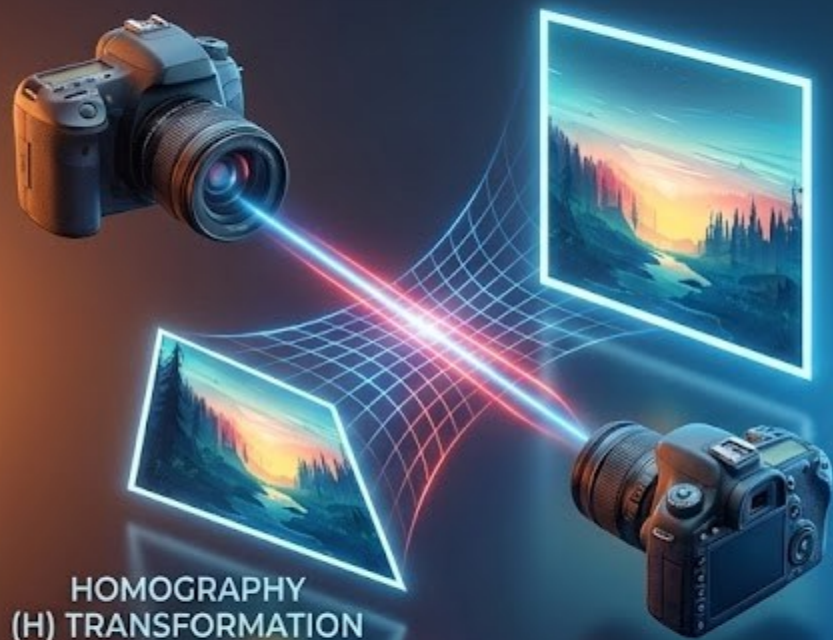
Key Mechanics:

- **Triangulation:** Once we match a point in both images, we can draw imaginary lines from both cameras through that point. Where the lines intersect in 3D space is the actual depth (distance) of the object.
- The **Fundamental Matrix** (F): Just like H is the key matrix for planes, F is the key matrix here. It encodes the relative position and orientation of the two cameras.

Real-World Applications::

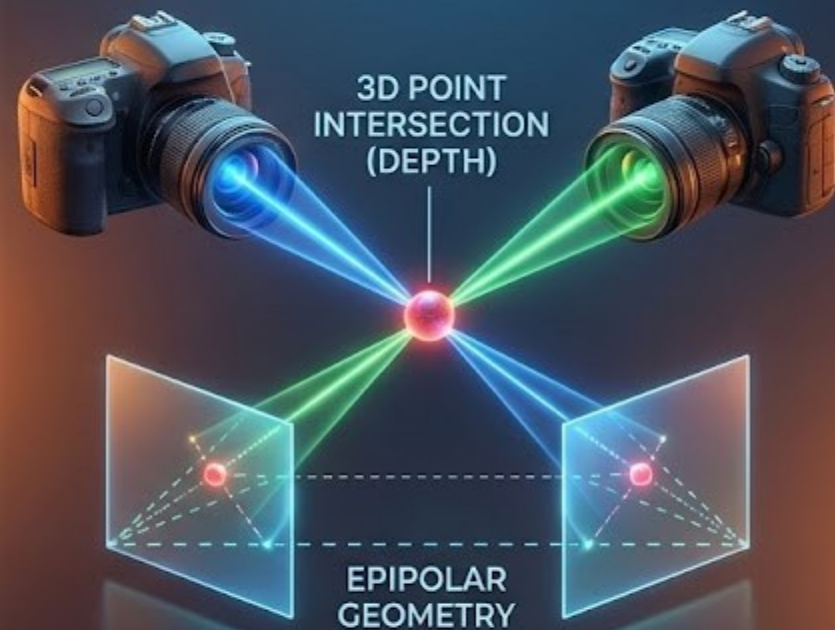
- **3D Reconstruction:** Turning a video of a statue into a 3D printable model.
- **Autonomous Driving:** Cars use stereo cameras to calculate how far away a pedestrian or another car is.
- **Augmented Reality (AR):** Phones use this to understand the 3D structure of a room to place virtual furniture on the floor.

SLIDE 1: HOMOGRAPHIES (PLANE WARPING)



RELATES IMAGES OF
A FLAT SURFACE.

SLIDE 2: MULTIPLE VIEW GEOMETRY (DEPTH FINDING)

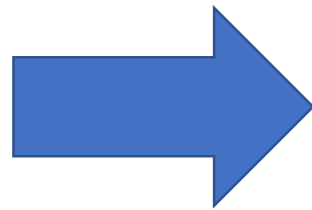


CALCULATES 3D STRUCTURE
FROM DIFFERENT VIEWS.

Basic Pinhole Projection (Metric)

-A simple case to understand how depth (Z) affects object size.

- Question: A camera with a focal length of 25 mm is used to photograph a person who is 1.8 meters tall. If the person is standing 5 meters away from the camera, what will be the height of the person in the image (on the sensor)?



$$\frac{y}{f} = \frac{Y}{Z}$$

Basic Pinhole Projection (Metric)

$$Y = 1.8 \text{ m} (\textit{Height of object})$$

$$Z = 5 \text{ m} (\textit{Distance / Depth})$$

$$f = 25 \text{ mm} = 0.025 \text{ m}$$

Basic Pinhole Projection (Metric)

$$y = f \times \frac{Y}{Z} = 0.025 \times \frac{1.8}{5}$$
$$= 0.009 \text{ m} = \mathbf{9 \text{ mm}}$$

Basic Pinhole Projection (Metric)

-A digital camera has the following specifications:

focal length $f = 800$ pixels.

Find the pixel coordinates (u, v) for a 3D point P

located at $(X, Y, Z) = (2, 1, 10)$

in the camera's coordinate system.